

1) Explain predefined functions of strings.

1. *strlen()*:

- Function: `size_t strlen(const char *str);`
- Purpose: Returns the length of the string `str` , excluding the null character (`\0`).
- Example: `strlen("Hello")` returns 5.

2. *strcpy()*:

- Function: `char *strcpy(char *dest, const char *src);`
- Purpose: Copies the string `src` to `dest` , including the null character (`\0`).
- Example: `strcpy(dest, "Hello")` copies "Hello" to `dest` .

3. *strncpy()*:

- Function: `char *strncpy(char *dest, const char *src, size_t n);`
- Purpose: Copies at most `n` characters from `src` to `dest` . If `src` is shorter than `n` , `dest` is padded with null characters (`\0`).
- Example: `strncpy(dest, "Hello", 3)` copies "Hel" to `dest` .

4. *strcat()*:

- Function: `char *strcat(char *dest, const char *src);`
- Purpose: Appends `src` to the end of `dest` , overwriting the null character (`\0`) at the end of `dest` .
- Example: `strcat("Hello ", "World")` returns "Hello World".

5. *strncat()*:

- Function: `char *strncat(char *dest, const char *src, size_t n);`
- Purpose: Appends at most `n` characters from `src` to `dest` , appending a null character (`\0`) at the end.
- Example: `strncat("Hello ", "World", 3)` returns "Hello Wor".

6. *strcmp()*:

- Function: `int strcmp(const char *s1, const char *s2);`
- Purpose: Compares `s1` and `s2` lexicographically. Returns:
 - 0 if `s1` is equal to `s2`
 - < 0 if `s1` is less than `s2`

- > 0 if $s1$ is greater than $s2$
- Example: `strcmp("Hello", "World")` returns a negative value.

7. `strcmp()`:

- *Function:* `int strcmp(const char *s1, const char *s2);`
- *Purpose:* Compares $s1$ and $s2$ lexicographically.
- *Return Values:*
 - 0 if $s1$ is equal to $s2$
 - < 0 if $s1$ is less than $s2$
 - > 0 if $s1$ is greater than $s2$

2) Write a code to print vowels ,constants,digits, white spaces using strings without using predefined functions

include

```
int main() { char str[100]; int vowels = 0, consonants = 0, digits = 0, spaces = 0; int i;
```

```
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
```

```
    for (i = 0; str[i]; i++) {
        if ((str[i] >= 'a' && str[i] <= 'z') || (str[i] >= 'A' && str[i] <= 'Z')) {
            switch (str[i]) {
                case 'a':
                case 'e':
                case 'i':
                case 'o':
                case 'u':
                case 'A':
                case 'E':
                case 'I':
                case 'O':
                case 'U':
                    vowels++;
                    break;
                default:
                    consonants++;
            }
        } else if (str[i] >= '0' && str[i] <= '9') {
            digits++;
        } else if (str[i] == ' ' || str[i] == '\t' || str[i] == '\n') {
            spaces++;
        }
    }
}
```

```
    }  
}  
  
printf("Vowels: %d\n", vowels);  
printf("Consonants: %d\n", consonants);  
printf("Digits: %d\n", digits);  
printf("White spaces: %d\n", spaces);  
  
return 0;  
  
}
```

Enter a string: Hello World 123 Vowels: 3 Consonants: 7 Digits: 3 White spaces: 1